

From: Lowe, Claire (EGLE)
Sent: 3/9/2026 2:58:31 PM
To: "Elaine J. Venema" <evenema@fveng.com>; "Jeff Dietlin" <utilities@cadillac-mi.net>
Subject: Cadillac WWTP - MAHL 2026
Attachments: Cadillac WWTP MAHL comments_Jan 2026 v1.docx

Jeff and Elaine,

Please see the attached comments regarding the January 2026 local limits evaluation submitted for phenol, beryllium, benzidine, and bromomethane. Please submit a revised MAHL and proposed local limits based on the attached comments by April 6, 2026. Please let me know if you have any questions regarding the comments or would like to discuss them further.

Thank you,

Claire Lowe

Environmental Quality Analyst
Water Resources Division, Gaylord District Office
Michigan Department of Environment, Great Lakes, and Energy
[989-330-4639](tel:989-330-4639) | LoweC9@Michigan.gov
Michigan.gov/EGLE



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY



ATTACHMENT NAME:

Cadillac WWTP MAHL comments_Jan 2026 v1.docx

ATTACHMENT TYPE:

Microsoft Office 12+, Office Open XML (OOXML/OpenXML): document (DOCX)

Cadillac WWTP MAHL, submitted January 2026

WRD Comments March 9, 2026. Claire Lowe.

- The study uses a plant removal efficiency (R_POTW) for beryllium of 31.5% as stated on page 5, Table 3, and calculated in Appendix A, page 20, Tables A13 and A14. The data presented show that all influent and effluent samples analyzed for beryllium were non-detect at <4.0 ug/L. It is unclear based on the data provided how a removal efficiency of 31.5% was calculated.
- Based on review of Appendix A, page 20, Tables A13 and A14, it appears that plant removal efficiency (R_POTW) has been calculated based on data which has been paired according to the average daily removal efficiency (ADRE) method. However, the dates provided in Tables A13 and A14, and the lab reports provided in Appendix B, do not indicate that the effluent sample collection is lagged relative to the influent sample collection. The ADRE method requires that paired samples are lagged to calculate accurate removal efficiency. In cases where samples are not collected as lagged pairs the mean removal efficiency (MRE) method may be applied. Use of the MRE method produces different removal efficiencies than those presented in Table 3, Table A14, and in subsequent calculations.
- Based on review of Appendix A, page 21, Table A15, it appears that removal efficiency for the primary effluent (R_prim) is also based on paired data that is not lagged. The ADRE method may not be applied when the data is not collected as lagged pairs. The MRE method may be applied to calculate the primary removal efficiency.
- Review of the Appendix A, page 20, Tables A13 and A14, and the lab reports in Appendix B, showed that there is data presented in Tables A13 and A14 for March 10, 2025, for which there is no corresponding lab report in Appendix B. Please provide the analytical data with the report. There is also a lab report for December 14, 2025, included in Appendix B, which includes influent and effluent data that is not presented in Tables A13 and A14, and not included in the removal efficiency calculations. If relevant data is not included in the calculations, please provide justification for excluding it.
- Based on review of the data presented in Appendix A, page 20, Tables A13 and A14, and the lab reports in Appendix B, removal efficiency for phenol, beryllium, and bromomethane is based entirely on non-detect results.

Phenol

The phenol limit is being evaluated as there is an existing local limit which may no longer be appropriate. Review of the effluent sample results for the Significant Industrial Users (SIUs) presented under Appendix A, pages 22 and 23 shows that several SIUs do discharge phenol above the limit of detection, and therefore it is appropriate to retain a local limit. As site specific removal can't be calculated based on the non-detect influent and effluent data the POTW should consider using the literature removal efficiency value available for phenol, found under Appendix R of the EPA local Limits Development Guidance.

Beryllium

The need for a local limit for beryllium is being evaluated because the NPDES permit contains an effluent limit for this parameter. While the influent and effluent data are all non-detect for beryllium, the NPDES permit includes an effluent limitation due to a previous detection at 8.1 ug/l in the effluent additional monitoring requirements (AMR) data which exceeds the chronic water quality value. Therefore, it appears appropriate to develop a local limit. As noted in section 3.0 of the MAHL report, literature values for beryllium are not available under Appendix R of the EPA Local Limits Development Guidance. However, a literature value for POTW removal is available under Section 11, Pollutant Loadings, Table 11-3, of the Development Document for Final Effluent Limitations Guidelines and Standards for the Iron and Steel Manufacturing Point Source Category (EPA-821-R-02-004). The POTW may use this beryllium literature removal efficiency value or may elect to use a surrogate value for the non-detect values (e.g. assume detection limit for influent data and ½ of the detection limit for the effluent sample).

Bromomethane

The City of Cadillac elected to evaluate the need for a local limit because the NPDES permit includes a report only requirement for bromomethane. All influent, effluent, SIU effluent, and background data provided for bromomethane were non-detect. It was noted during review of this data that bromomethane was added to the NPDES permit as a report only parameter because monitoring conducted under the AMR requirements of the NPDES permit did not have a sufficiently low detection limit to determine if there was reasonable potential to violate the water quality standards. Monitoring conducted under the current

NPDES permit is also not of a sufficiently low detection limit to determine compliance with the effluent limitation. Development of a bromomethane local limit is not required under the current NPDES permit and is not recommended based on the data available. Cadillac WWTP must implement bromomethane monitoring with a quantification limit which meets the requirements of the NPDES permit under Part I, Section A.2. Monitoring which meets this requirement will determine if there is a reasonable potential to violate the water quality standard for bromomethane and therefore if an effluent limit will be included in future NPDES permits. A local limit may be unnecessary for bromomethane and this can't be determined based on the data provided.

- The domestic and commercial background flow (Q_{dom}) included in Appendix A, Tables A5, A6, and A7 (pages 14 – 16) is stated as 0.8571 MGD. This does not agree with other areas of the report and calculations where Q_{dom} is reported as 0.863 MGD.